DSA1604 - Knowledge Engineering and Data Science

Activity 1: Knowledge Detective

Topic: Knowledge Engineering

Questions:

1. Student Attendance Prediction: What information is required to predict whether a student may have low attendance?

2. Medical Diagnosis: What symptoms and medical knowledge are needed to help identify a possible illness?

3. Library Management: What knowledge is required to determine whether a book can be issued to a student?

4. Computer Troubleshooting: What facts are needed to identify why a computer is not connecting to the internet?

5. Smart Campus Management: What knowledge is required to identify an available classroom for a scheduled lecture?

Expected student output

For each question, students should identify:

- Facts: Information needed to understand the problem.
- Rules: Conditions used to make decisions.
- Expert knowledge: Experience or guidelines needed to solve the problem

Activity 2: Build a Semantic Network

Topic: Knowledge Representation

1. Draw a semantic network using the concepts Student, Course, Teacher, and Department. Represent each concept as a node and each relationship as a labeled arrow.

2. Draw a semantic network using the concepts Doctor, Patient, Hospital, and Medicine. Show the meaningful relationships between these concepts using labeled arrows.

3. Draw a semantic network using the concepts Book, Author, Library, and Student. Connect the concepts using appropriate relationships.

4. Draw a semantic network using the concepts Employee, Manager, Company, and Project. Represent their relationships using labeled arrows.

5. Draw a semantic network using the concepts Customer, Product, Order, and Payment. Show how these concepts are connected in an online shopping system.

Activity 3: IF–THEN Rule Challenge – 5 Questions

Topic: Knowledge-Based Systems and Production Rules

1. Write an IF–THEN rule to identify a student whose attendance is below the required threshold.

2. Write an IF–THEN rule to determine whether a library book can be issued to a student based on its availability.

3. Write an IF–THEN rule to identify whether a student is eligible to register for an examination based on attendance and academic requirements.

4. Given a student with 65% attendance, apply an appropriate IF–THEN rule using an assumed minimum attendance requirement of 75%. State the decision made by the knowledge-based system.

5. Write an IF–THEN rule to determine whether a patient should receive an appointment based on doctor availability and the patient's urgency level.

6. Write an IF–THEN rule to determine whether a vehicle is allowed to enter the campus parking area based on parking space availability and valid vehicle registration.

Expected output: Students should write production rules in the form IF condition THEN action and apply them to the given scenarios to determine the outcomes.

ANSWERS

Activity 1: Knowledge Detective

Topic: Knowledge Engineering

1. Student Attendance Prediction

- **Facts:** Previous attendance, class schedule, absences, late arrivals, holidays, and student participation.
- **Rules:** If attendance is below the required percentage or frequent absences occur, predict low attendance.
- **Expert Knowledge:** Attendance policies and factors that commonly affect student attendance.

2. Medical Diagnosis

- **Facts:** Symptoms, age, medical history, temperature, test results, and medications.
- **Rules:** If specific symptoms match a disease pattern, identify the possible illness.
- **Expert Knowledge:** Medical guidelines, disease symptoms, and doctor experience.

3. Library Management

- **Facts:** Book availability, student's library status, borrowing limit, and due books.
- **Rules:** If the book is available and the student has not reached the borrowing limit, the book can be issued.
- **Expert Knowledge:** Library rules, borrowing policies, and reservation guidelines.

4. Computer Troubleshooting

- **Facts:** Wi-Fi status, network connection, IP address, router status, and network settings.
- **Rules:** If Wi-Fi is connected but there is no internet, check the router or network configuration.
- **Expert Knowledge:** Networking concepts and common troubleshooting procedures.

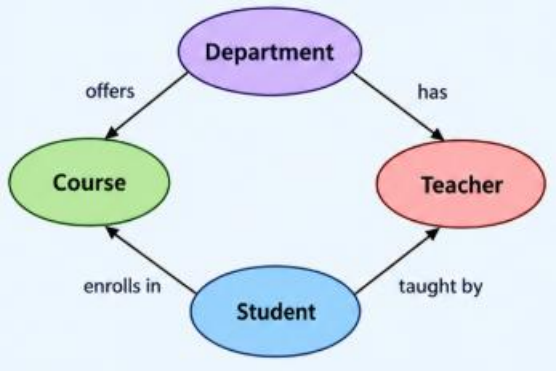
5. Smart Campus Management

- **Facts:** Classroom availability, timetable, room capacity, equipment, and current bookings.
- **Rules:** If a classroom is free at the required time and has sufficient capacity, it can be assigned.
- **Expert Knowledge:** Timetable rules, classroom capacity, and scheduling requirements.

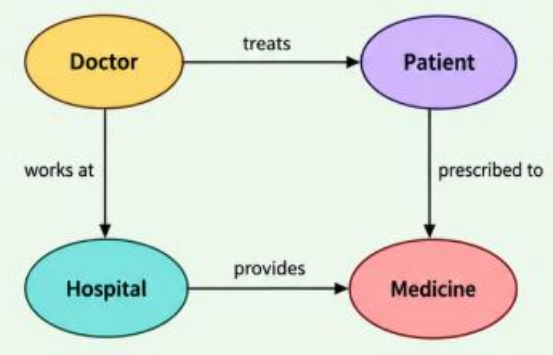
Activity 2: Build a Semantic Network

Topic: Knowledge Representation

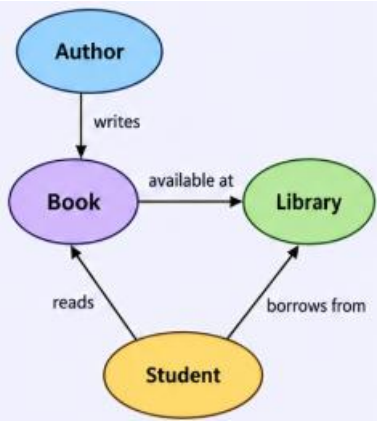
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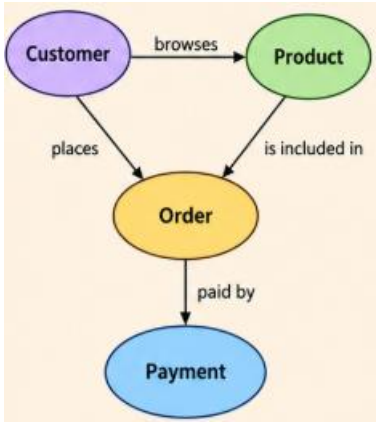
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Activity 3: IF–THEN Rule Challenge

Topic: Knowledge-Based Systems and Production Rules

1. Low Attendance

IF student attendance is below 75%
THEN identify the student as having low attendance.

2. Library Book Issue

IF the book is available AND the student has not reached the borrowing limit
THEN issue the book to the student.

3. Examination Registration

IF attendance is at least 75% AND academic requirements are satisfied
THEN student is eligible to register for the examination.

4. Student with 65% Attendance

IF attendance is below 75%
THEN student is not eligible based on the attendance requirement.

Given: Attendance = 65%

Decision: The student is not eligible because 65% is below the required 75%.

5. Patient Appointment

IF doctor is available AND patient urgency is high
THEN provide an immediate appointment.

IF doctor is available AND patient urgency is normal
THEN provide a regular appointment.

IF doctor is unavailable
THEN schedule the patient for the next available time.

6. Campus Parking

IF parking space is available AND vehicle registration is valid
THEN allow the vehicle to enter the campus parking area.

IF parking space is unavailable OR vehicle registration is invalid
THEN deny entry.